Sturmian sequences of $\sqrt{n}$

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1 The window count

Consider the sequence generated by replacing every 1 by 10 and every 0 by 1. Start at 1 and repeat:

$$1, \quad 10, \quad 101, \quad 10110, \quad 10110101, \quad 1011010110110, \quad \ldots$$

The limit is the *Fibonacci sequence*

$$\ell \;=\; 1\,0\,1\,1\,0\,1\,0\,1\,1\,0\,1\,1\,0\,1\,0\,1\cdots$$

Slide a window m letters wide along it and write down every distinct block. Let $F(m)$ be how many there are. For $m = 3$:

position	window	block	
0	101101011011010101	101	new
1	101101011011010101	011	new
2	101101011011010101	110	new
3	101101011011010101	101	seen
4	101101011011010101	010	new
5	101101011011010101	101	seen
6	101101011011010101	011	seen
7	101101011011010101	110	seen
8	101101011011010101	101	seen
$\cdots$ and nothing new appears after that $\cdots$			

Four distinct blocks, so $F(3) = 4$.

Definition 1. A sequence is *Sturmian* when $F(m) = m + 1$ for every $m \geq 1$.

The Fibonacci sequence is Sturmian.

2 Morse and Hedlund

There is an alternative way to characterize Sturmian sequences.

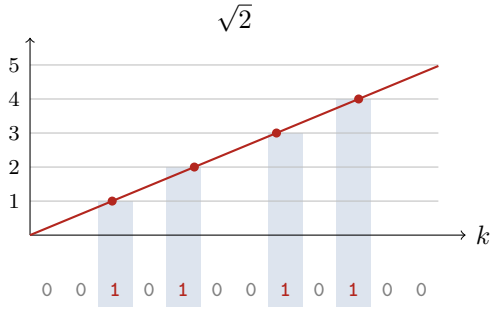
Theorem 2 (Morse and Hedlund)**.** *A sequence is Sturmian if and only if there is an irrational* $\alpha \in (0,1)$ *and a* $\rho \in [0,1)$ *with*

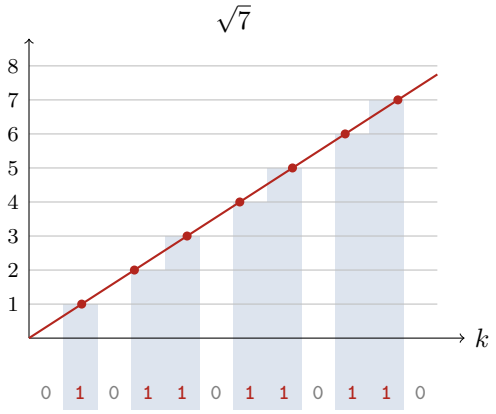
$$\ell_k \;=\; \lfloor (k+1)\alpha + \rho \rfloor - \lfloor k\alpha + \rho \rfloor.$$

The formula counts integer crossings. Walk along the real line in steps of α, so that after k steps you stand at position $k\alpha + \rho$. Each step is shorter than a whole unit, so it passes through at most one integer. We record $\ell_k = 1$ when the step from k to $k+1$ passes an integer and $\ell_k = 0$ otherwise.

The equivalence of these two definitions is not obvious.

After k steps you have travelled a distance of $k\alpha$. So about $k\alpha$ of the first k letters are 1s, and the proportion of 1s is α. We call this both the density and the slope. The second number ρ tells us only where we start reading the sequence.





3 The sequence of a square root

Put $a_0 = \lfloor \sqrt{n} \rfloor$ and take the slope to be

$$\alpha \;=\; \sqrt{n} - a_0, \qquad \ell_k \;=\; \lfloor (k+1)\alpha \rfloor - \lfloor k\alpha \rfloor.$$

If $\sqrt{n} = [\, a_0;\ \overline{a_1, \ldots, a_L}\,]$ then $\alpha = [\, 0;\ \overline{a_1, \ldots, a_L}\,]$, periodic from a_1.

$\alpha = [\, 0;\ a_1, a_2, a_3, \ldots\,]$ means $\alpha = 1/(a_1 + \beta)$ with $\beta = [\, 0;\ a_2, a_3, \ldots\,]$, so

$$\frac{1}{\alpha} \;=\; a_1 + \beta, \qquad \beta \in (0, 1).$$

From which it follows that:

$$a_1\alpha \;<\; 1 \;<\; (a_1 + 1)\alpha.$$

The next 1 therefore comes after a_1 or $a_1 + 1$ steps, and the number of consecutive 0s between any two consecutive 1s must be $a_1 - 1$ or a_1.

We define a new sequence ℓ', the *derived sequence*, by setting

$$\ell'_k \;=\; \begin{cases} 1 & \text{if } a_1 \text{ zeros lie between the } k\text{th and } (k{+}1)\text{th 1 of } \ell, \\ 0 & \text{if } a_1 - 1 \text{ do.} \end{cases}$$

4 $n = 2$

$\sqrt{2} = [\, 1;\ \overline{2}\,]$, so $L = 1$ and $\alpha = \sqrt{2} - 1$.

$$\ell \;=\; 00101001010010101001010010101001010010101001010010\cdots$$

The 1s sit at

$$2,\ 4,\ 7,\ 9,\ 12,\ 14,\ 16,\ 19,\ 21,\ 24,\ 26,\ 28,\ 31,\ 33,\ \ldots$$

and the numbers of 0s between them are

$$1,\ 2,\ 1,\ 2,\ 1,\ 1,\ 2,\ 1,\ 2,\ 1,\ 1,\ 2,\ 1,\ 2,\ \ldots$$

The derived sequence is thus:

$$\ell' \;=\; 0101001010010101010\cdots$$

Observe that the derived sequence is the original sequence with the first letter removed.

Here

$$\frac{1}{\alpha} \;=\; \sqrt{2} + 1 \;=\; 2 + \beta, \qquad \beta \;=\; \sqrt{2} - 1 \;=\; \alpha,$$

5 $n = 7$

$\sqrt{7} = [2;\ \overline{1,1,1,4}]$, so $L = 4$ and $\alpha = \sqrt{7} - 2$.

$$\ell \ = \ 0101101101101101011011011011010110101101101\cdots$$

The 1s sit at

$$1,\ 3,\ 4,\ 6,\ 7,\ 9,\ 10,\ 12,\ 13,\ 15,\ 17,\ 18,\ 20,\ 21,\ \ldots$$

and the numbers of 0s between them are

$$1,\ 0,\ 1,\ 0,\ 1,\ 0,\ 1,\ 0,\ 1,\ 1,\ 0,\ 1,\ 0,\ 1,\ \ldots$$

The blocks are 1 and 10:

ℓ	0	10	1	10	1	10	1	10	1	10	10	1
ℓ'		1	0	1	0	1	0	1	0	1	1	0

Four derivations run through the period:

	α		sequence
ℓ	$\sqrt{7} - 2$	$[0;\ \overline{1,1,1,4}]$	0101101101101101011011011011010110
derive	$(\sqrt{7} - 1)/3$	$[0;\ \overline{1,1,4,1}]$	1010101011010101010110101010101101
derive^2	$(\sqrt{7} - 1)/2$	$[0;\ \overline{1,4,1,1}]$	1111011111011110111111011111101111
derive^3	$(\sqrt{7} - 2)/3$	$[0;\ \overline{4,1,1,1}]$	0001000010001000010000100010000001
derive^4	$\sqrt{7} - 2$	$[0;\ \overline{1,1,1,4}]$	1011011011011010110110110110110101

The last row is ℓ shifted one letter. Every sequence in the table is Sturmian.

6 The Fibonacci sequence

$$\ell \ = \ 1011010110110101101011011010110110101101011010110\cdots$$

The 1s sit at

$$0,\ 2,\ 3,\ 5,\ 7,\ 8,\ 10,\ 11,\ 13,\ 15,\ 16,\ 18,\ 20,\ 21,\ \ldots$$

and the numbers of 0s between them are

$$1,\ 0,\ 1,\ 1,\ 0,\ 1,\ 0,\ 1,\ 1,\ 0,\ 1,\ 1,\ 0,\ 1,\ \ldots$$

The blocks are 1 and 10:

ℓ	10	1	10	10	1	10	1	10	10	1	10	10
ℓ'	1	0	1	1	0	1	0	1	1	0	1	1

Observe that the derived sequence is the original sequence.

Take that as the defining property and solve for α. The steps are 1 and 2, so $a_1 = 1$ and $1/\alpha = 1 + \beta$. The derived sequence is ℓ, so $\beta = \alpha$ and

$$\frac{1}{\alpha} \ = \ 1 + \alpha, \qquad \alpha^2 + \alpha - 1 \ = \ 0, \qquad \alpha \ = \ \frac{\sqrt{5} - 1}{2} \ = \ \frac{1}{\varphi}.$$

The golden mean falls out as the slope of the Fibonacci sequence.

In continued fractions,

$$\frac{1}{\varphi} \ = \ [0;\ 1, 1, 1, 1, \ldots],$$

because $\varphi = 1 + 1/\varphi$. Deriving deletes a_1 and leaves the list unchanged.

7 Square roots as sequences

Deriving a Sturmian sequence sends the slope α to $\{1/\alpha\}$, which deletes the first term a_1 of the continued fraction expansion. So the tower of derived sequences circles back exactly when the continued fraction of α repeats.

Theorem 3 (Lagrange)**.** *The continued fraction of α is eventually periodic if and only if α is a quadratic irrational,*

$$\alpha = \frac{a + \sqrt{n}}{b},$$

with a, b and n integers and n not a square.

Sturmian sequences with cyclical derived sequences give us another way to think about quadratic irrationals.